CS554 Midterm #2

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I. Question

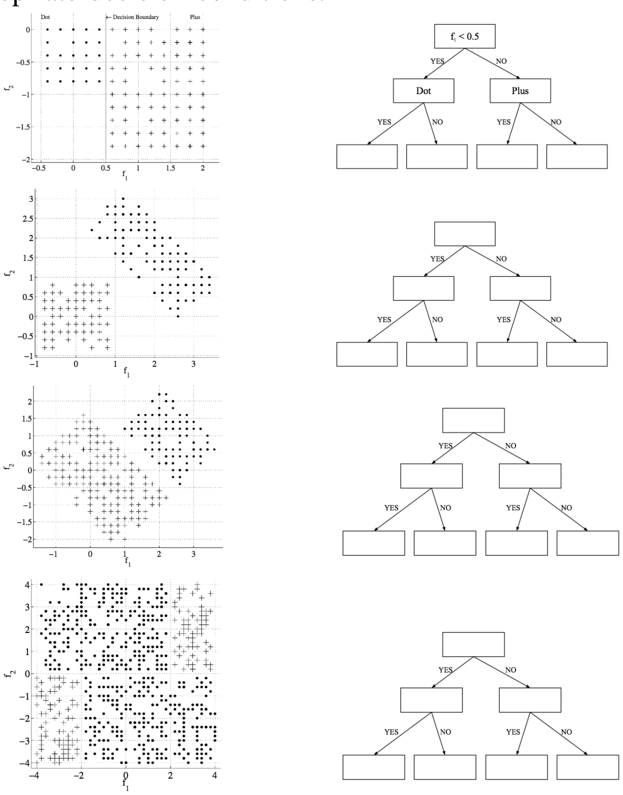
Run the linear perceptron algorithm one epoch for the following data:

Feature 1	Feature 2	Class
1	1	1
1	0	1
0	1	1
0	0	0

Assume that the initial weights are all 0.1, and the learning rate is 0.1. Do batch update, i.e. calculate the update of the weights for all data at once.

II. Question

For the following data, fill the decision nodes with appropriate decision conditions.



III. Question

Let us say we have 4 states:
S_1: A, S_2: C, S_3: G, S_4: T

with initial probabilities: $\mathbf{\Pi}$ = [0.3, 0.4, 0.1, 0.2]
and the transition matrix is:

$$\mathbf{A} = \left| \begin{array}{cccc} 0.2 & 0.2 & 0.3 & 0.3 \\ 0.1 & 0.3 & 0.2 & 0.4 \\ 0.4 & 0.4 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.7 \end{array} \right|$$

- Calculate the probability of the sequence ACCGTC.
- Given twelve sequences AAGCT, TATTA, GTAGT, TTATT, GAGCT, AAAAT, ACCTT, CTTAA, TGATT, GGTAA, TACAG, CGAAC estimate the initial probabilities for $\mathbf{\Pi}$ and $\mathbf{A}$.

IV. Question

Propose a tree induction method where each split in each node is in the form of $a \leq x_i \leq b$.

V. Question

Consider the alternative error function in linear perceptron for regression problem:

$$E = \frac{1}{2}(r - y)^2 + \alpha \sum_i w_i^2 \quad (1)$$

where N represents the number of instances, r represents the actual output, y represents the calculated output, v_i are the weights between the input units and the output unit. Derive the gradient descent update rules for the weights.

VI. Question

If each base-learner is independently and identically distributed and correct with probability $p >$ 1/2, what is the probability that a majority vote over L classifiers gives the correct answer? Calculate the probability for L = 5 and p = 3 / 4.

VII. Question

Consider the three learners d_1, d_2, and d_3; and the posterior probabilities found for three classes for those learners.

	C_1	C_2	C_3
d_1	0.2	0.5	0.3
d_2	0.1	0.6	0.3
d_3	0.3	0.3	0.4

Calculate the posterior values for three classes for sum, median, minimum, maximum and product rules.